

Project Title

E-Commerce Product Entity-Resolution & Matching Platform

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

Online shopping platforms often contain multiple listings of the same product from different sellers. These listings may have different product names, descriptions, prices, or formatting, making it difficult for customers to identify the same product and compare prices. The E-Commerce Product Entity-Resolution & Matching Platform is designed to solve this problem by automatically detecting similar product listings and grouping them into a single product. This helps users easily compare prices offered by different sellers and make better purchasing decisions while reducing duplicate product records.

The proposed system is developed using the MERN Stack, which includes MongoDB, Express.js, React.js, and Node.js. It uses several Data Structures and Algorithms to perform efficient product matching. Product titles are first cleaned and tokenized, after which an Inverted Index is created to reduce unnecessary comparisons. The similarity between product titles is calculated using Jaccard Similarity, while Levenshtein Distance is used to identify minor spelling differences or typing errors. A weighted similarity score is then used to determine whether two products represent the same item, and the Union-Find (Disjoint Set Union) algorithm groups matching products into a single cluster.

The platform provides features such as product upload, product matching, duplicate detection, price comparison, and grouped product display. By combining modern web technologies with efficient algorithms, the system offers a practical solution for e-commerce applications while demonstrating the real-world use of data structures and algorithms. The project is scalable, user-friendly, and can be enhanced in the future with advanced AI-based product matching, recommendation systems, and real-time data integration from multiple online marketplaces.

Signature of the Guide